



BUILD A WORLD MODEL FROM SCRATCH · LECTURE 1

What Is a *World Model*, Really?

Four ideas, built carefully: the agent–environment loop, states vs observations, eighty years of history, and the map of the whole field.

THE LOOP

STATE $\neq$ OBSERVATION

80 YEARS

THE TAXONOMY

Vizuara AI Labs · Build a World Model from Scratch

Four ideas – that's the whole lecture

1 · The agent–environment interface

One loop connects every AI system that acts: agent → action → environment → observation. Four words, one picture.

2 · States vs observations

The world's full truth vs the part you get to see. The single distinction this whole field is built on.

3 · Eighty years of history

World models are not new – 1943 to today, in five steps.

4 · The taxonomy: Renderer / Simulator / Planner

Three questions that organise every world-model paper ever written.

Explanation from Rajat

— WHITEBOARD —



IDEA 1 OF 4

The Agent–Environment Interface

Every acting AI system — a game bot, a robot, a self-driving car
— talks to its world through one identical loop.

AGENT • ACTION • ENVIRONMENT • OBSERVATION

One loop, four words

Action a

What the agent does. Its only way to influence the world.

State s

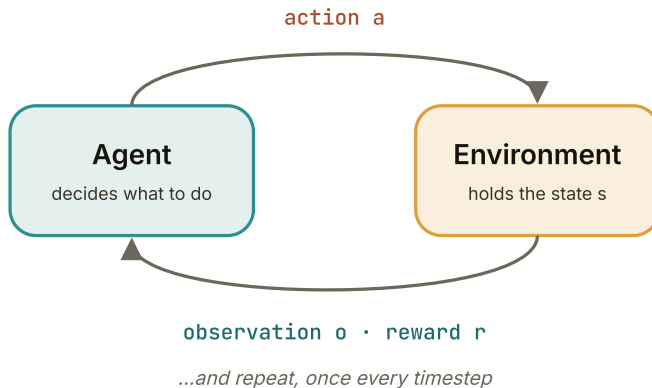
The world's complete internal situation. Lives *inside* the environment.

Observation o

What the environment shows back. What the agent actually receives.

Reward r

A score for how good that step was. Optional — not every system uses one.



Example: one timestep of a platformer



Press → · game computes · new frame comes back. The whole interface in one move.

1 The agent presses →

That's the action. One button, sent to the game.

2 The game updates its state

Internally: the character's position moves right, momentum changes, an enemy advances one step. All invisible bookkeeping.

3 The game returns a new frame

That's the observation — a picture of the new situation. Plus a reward: +1 if a coin was collected.

4 Repeat

60 times a second, for the whole game. This loop *is* playing.

So what is a world model? A learned stand-in for the environment.

The environment does two jobs every timestep. A **world model** is a neural network trained to do those same two jobs:

Job 1 — update the situation

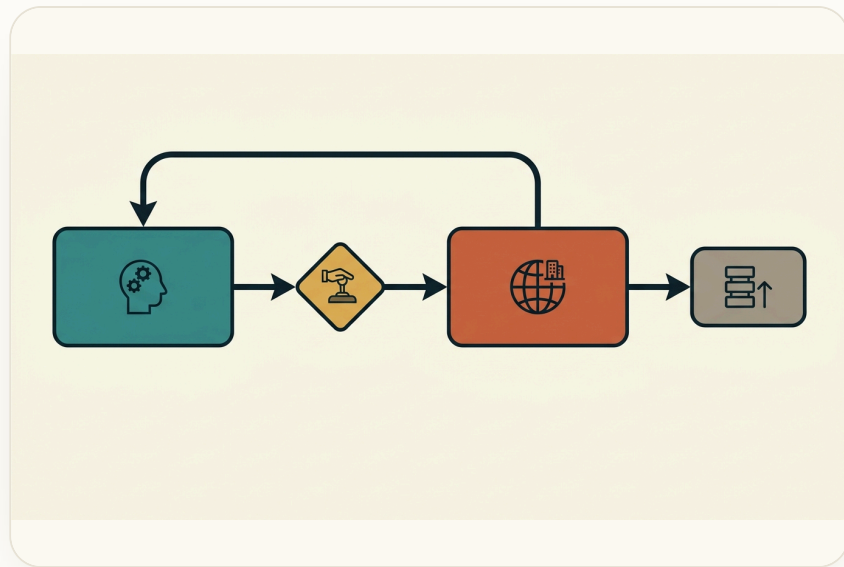
Given where things are and what the agent did → where things are next.
The rules of the world.

Job 2 — show something

Given the new situation → produce what the agent would see.

Why bother?

Because if a network can do both jobs, the agent can *unplug the real world* and keep practising inside the network — thousands of times faster, at zero cost, with zero risk. That is the entire promise of this field.



The world model learns to play the part of the orange box.

Explanation from Rajat

— WHITEBOARD —



IDEA 2 OF 4

State vs Observation

The world's full truth, versus the part of it you get to see.
Everything hard about world models starts right here.

THE SINGLE MOST IMPORTANT DISTINCTION IN THIS SERIES

Two definitions

State s

the world's complete truth

Everything that determines what happens next. Every position, every velocity, every hidden variable. The world always knows its own state — **you usually don't**.

Driving in fog is the feeling of this distinction: the road ahead *exists* — its state is fully definite — but your observation ends a few metres out. And you still have to drive.

Observation o

your window onto it

What your sensors return — a camera frame, a game screen. A **partial, incomplete snapshot** of the state. Some truth survives; some is simply not there.



Chess vs poker — the cleanest possible contrast



Chess: you see everything

The board *is* the state. Nothing is hidden — what you observe and what determines the future are the same thing. **Fully observable.**



Poker: the crucial part is hidden

The state includes your opponent's cards — and your observation doesn't. You must act on incomplete truth. **Partially observable.**

Now the important question

Which one is the real world like? Cameras, microphones, robot sensors — is real life chess, or poker? **It is always poker.** No sensor ever captures the full state. That's why this distinction isn't a technicality — it's the default condition of every real agent.

Look at one frame of Pong. Which way is the ball moving?

The ball is perfectly visible. Nothing is blurry, nothing is blocked. And yet — **you cannot answer.**

- **The frame shows position**

Where the ball is: fully visible.

- **The frame cannot show velocity**

Where it's *going*: not in the image. Not hidden — *absent*. A single photo of a moving object contains no motion.

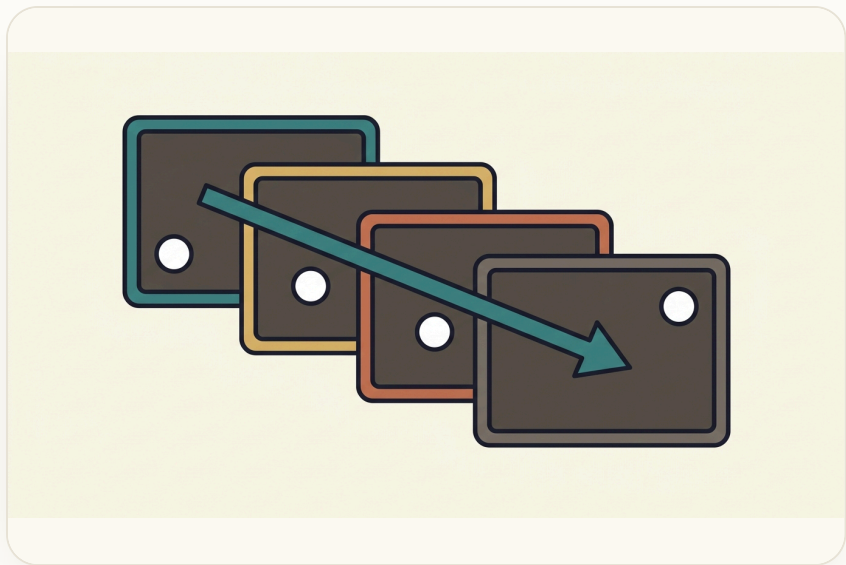
- **But velocity decides the future**

Ball moving left and ball moving right lead to totally different next frames. The one thing the future depends on is the one thing you can't see.



Position: visible. Velocity: gone. The observation is missing exactly what matters.

The fix: bring in time



Four frames, one direction. History gives back what the snapshot lost.

Show me **four consecutive frames** instead of one, and the answer appears instantly — the ball traces a line, and the line has a direction.

- **No single frame gained anything**

Each is still velocity-blind. The information came from *comparing* them — from history.

- **This is the master recipe**

What one observation is missing, a sequence of observations can recover. Every world model you'll meet in this series is, at heart, a machine for doing exactly this.

Two names to file away

MDP: observation = state (chess). **POMDP**: observation $\subset$ state (poker, Pong, real life). You'll see these acronyms in every paper — they mean nothing more than this slide.

Explanation from Rajat

— WHITEBOARD —



IDEA 3 OF 4

Eighty Years of World Models

The most common belief about world models is that they're a 2024 invention. Five stops through history say otherwise.

1943 → TODAY

Craik: the mind carries a small-scale model of reality



1943: the idea. The next eighty years: waiting for the hardware.

Cambridge psychologist Kenneth Craik, 1943 — five years before the transistor. His claim about the brain, paraphrased:

The claim

An organism carrying a small-scale model of the world can "try out alternatives, conclude which is best, and react to future situations *before they arise*."

Read it slowly: try alternatives → conclude the best → act before it happens. That's a complete description of what the systems in this series do — written as neuroscience, eighty years early.

Tolman: rats that learned a map, not a habit

Let rats wander a maze with **no food, no reward**. Then place food somewhere.

- **They found it far faster than fresh rats**

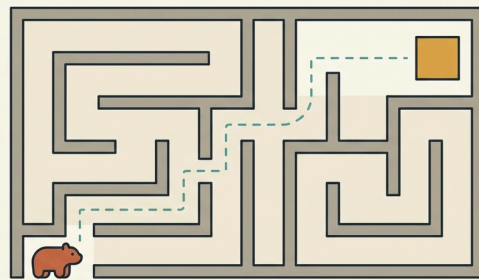
So the "pointless" wandering had taught them something — with zero reward involved.

- **And they took shortcuts they'd never walked**

Impossible if they'd only memorised rewarded turns. Possible only if they'd built a **map** — a model of the maze itself.

- **The lesson that still matters**

Learning the world and *learning what to do* are two different things — and the first can happen without any reward at all.

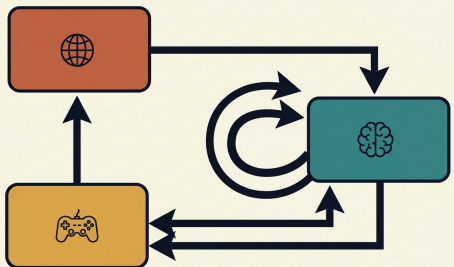


A shortcut never taken before: evidence of a map in the head.

Kalman flew it to the Moon. Sutton made it a training loop.



Predict, then correct.



World → model → policy — and around again.

1960 — the Kalman filter

How do you track a spacecraft's true state from noisy sensors?

Predict

where it should be, then

correct

with the measurement. Apollo ran on this. It's the first engineered answer to "state hidden, observation noisy" — our Idea 2, solved with 1960s math.

1991 — Sutton's Dyna

The first RL recipe with a learned model in the loop: act in the world → update your model of it → let the policy practise *inside the model*, many times per real step. Every modern system in this series still runs this exact loop.

Ha & Schmidhuber: "World Models" – the name and the proof

The 2018 paper that gave the field its name — and its most striking demonstration.

- **A neural network learned a racing game**

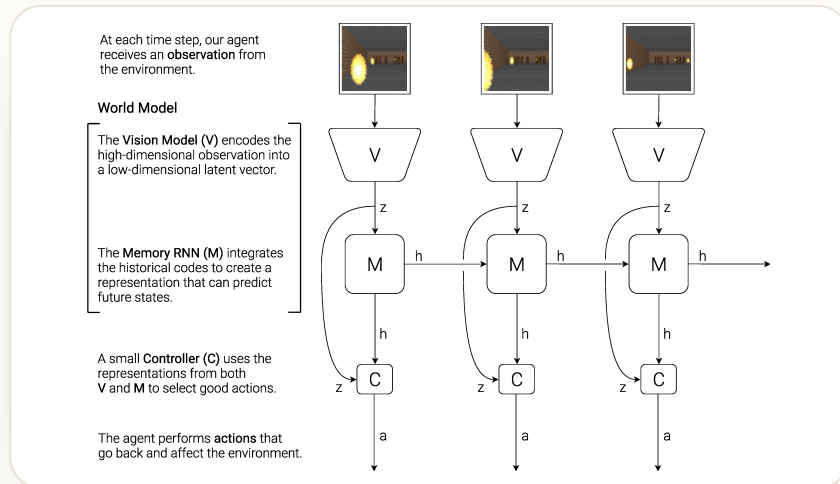
Not to *play* it — to *be* it: compress each frame, predict the next one. The two jobs from Idea 1, done by networks.

- **Then the agent trained inside the copy**

The policy learned to drive *entirely within the network's imagination* — the real game switched off.

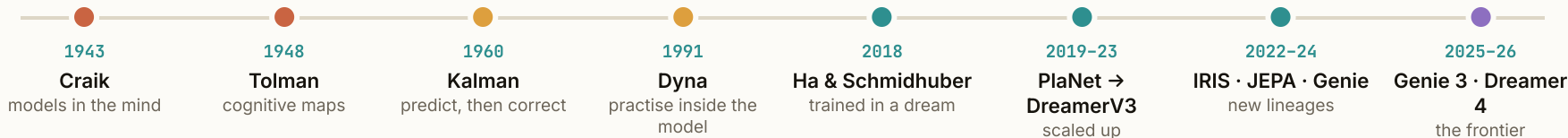
- **And it worked in the real game**

Skills learned in the dream transferred to reality. Craik's 1943 sentence, finally executed in code.



From the original paper. We'll rebuild this system ourselves later in the series.

Old idea. New compute. That's the whole story.



✗ Misconception, retired: "world models are new"

The idea is eighty years old. What's new is the ability to execute it — deep networks to learn from pixels, and GPUs to run millions of imagined steps. **The concept was waiting for the compute.**

Explanation from Rajat

— WHITEBOARD —



IDEA 4 OF 4

The Taxonomy: Renderer · Simulator · Planner

"World model" means different things in different papers. Three questions sort all of them.

THE MAP YOU'LL USE FOR THE WHOLE SERIES

Push a cup, and your brain answers three questions



One everyday gesture; the entire field hiding inside it.

Before my finger moves, I already know:

- **"What would I *see*?"**

A picture of the cup, slid over — an image of a moment that hasn't happened.

- **"What *happens* if I push harder?"**

It tips. Push at the rim — it spins. Different action, different outcome.

- **"What *should* I do?"**

To get it to the far coaster without spilling: two gentle pushes, not one hard one.

Three different questions — and each one defines a whole family of systems.

The three faces



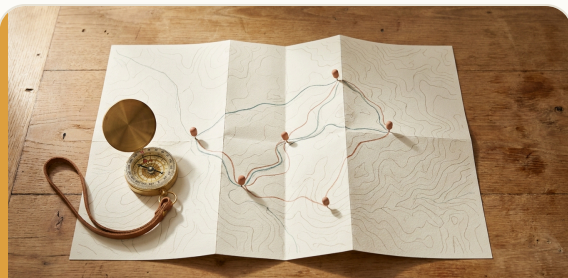
Renderer — "what would I see?"

Generates realistic observations. Judged on how it *looks*. Sora-class video models, Genie, Marble.



Simulator — "what happens if I act?"

Predicts consequences of actions. Judged on being *right*. Dreamer, IRIS, JEPA-style models.



Planner — "what should I do?"

Turns predictions into decisions. Judged on *reaching the goal*. MPC, search, RL-in-imagination.

How to use this map

When you meet any "world model" — a paper, a product, a headline — ask which question it answers. Every system in this series slots cleanly into one (sometimes two) of these three. That's the shelf we'll place everything on, lecture by lecture.

The map, applied to names you've heard

SYSTEM	WHAT IT DOES	FACE
Sora-class video models	beautiful clips from a prompt — no actions anywhere	RENDERER
Genie (DeepMind)	playable worlds — press a button, the frame responds	RENDERER + actions
Dreamer (DeepMind)	predicts consequences; agent trains inside its imagination	SIMULATOR
MuZero-style planning	searches over imagined futures to pick a move	PLANNER

One question to carry out of today

"Is it action-conditioned?" — does the output *change when the action changes*? A renderer can ignore actions; a simulator cannot. That single question separates pretty video from a true model of consequences.

✗ Misconception, retired: "a world model = a video generator"

Video generation is one face of three — and the only one that never needs to know what *you* did.

Explanation from Rajat

— WHITEBOARD —

Four things to keep

1 Everything that acts lives on one loop

Agent → action → environment → observation (+ reward). A world model is a neural network learning to play the environment's part.

2 The observation is not the state

Chess is the exception; poker is the rule. One Pong frame can't show velocity — and what one observation misses, a *sequence* can recover.

3 The idea is eighty years old

Craik → Tolman → Kalman → Dyna → Ha & Schmidhuber. New compute, not new concept.

4 Three questions organise the whole field

What would I see? What happens if I act? What should I do? — Renderer, Simulator, Planner.

Lecture 2 – The World Modeler's Toolkit

The background you need before the first build — delivered just in time, no detours.

- **Just enough RL**

Rewards over time, value, and the two classic ways to learn behaviour.

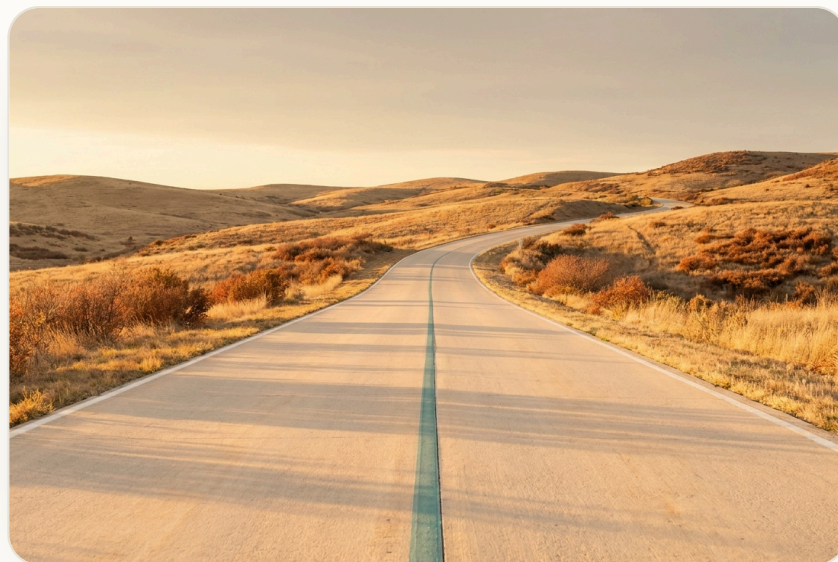
- **Just enough representation**

What it means to compress an image into a few numbers — the idea every world model is built on.

- **Then we start building**

From Lecture 3 onward, every lecture ends with code you can run.

●●● Vizura AI · Build a World Model from Scratch



See you in Lecture 2.